

UNDERSTANDING COSMOLOGY

EARLY 20TH-CENTURY ASTRONOMERS DEVELOPED A THEORY TO EXPLAIN THE ORIGIN OF THE UNIVERSE

Cosmology is the scientific study of the Universe as a whole—the term is derived from the Greek word for Universe, *cosmos*. Cosmologists are interested in how the Universe began, how it works (particularly on the largest scales), how it will develop in the future, and if and how it will end.

It was only in the 1920s that astronomers discovered other galaxies outside our own. Belgian astronomer and priest Georges Lemaître (1894–1966) applied the equations of Einstein’s General Relativity to the Universe as a whole; the results suggested that the Universe might be expanding. He proposed that if that were true, it must long ago have been very small, very dense, and very hot—a state he called the primordial atom.

IN THE BEGINNING
In 1929, American astronomer Edwin Hubble discovered that galaxies are moving away in every direction, suggesting that the Universe is indeed expanding—and that Lemaître’s theory might be correct. British astronomer Fred Hoyle rejected these ideas, but coined the term Big Bang in 1950 to help explain them. Big Bang theory remains a very likely explanation of how the Universe began.



GEORGES LEMAITRE
After a strict Jesuit upbringing, Lemaître studied civil engineering, then physics and mathematics. He was ordained as a Jesuit priest in 1923.

THE ORIGIN OF THE UNIVERSE
As the Universe expanded after the Big Bang, it also cooled. Some of its energy turned into fundamental particles, and the basic forces of nature came into existence.

DIAMETER	3×10^{-26} ft/ 10^{-26} m	33 ft/10 m	62 miles (10^9 m/100 km)
TEMPERATURE	10^{27} K (1,800 trillion trillion °F/100 trillion trillion °C)	10^{27} K (1,800 trillion trillion °F/100 trillion trillion °C)	10^{22} K (18 billion °F/100 billion billion °C)
	Cosmic inflation <i>At the moment of the the Big Bang, the entire Universe was much smaller than an atomic nucleus. Within a tiny fraction of a second, it underwent an inconceivably rapid expansion called cosmic inflation.</i>	Particle soup <i>When cosmic inflation ended, still well within the first second, the Universe was tiny and hot. Energy created pairs of particles and antiparticles, which pop into existence fleetingly before annihilating each other.</i>	Separation of forces <i>Originally, what we know as electromagnetism, gravity, and the weak and strong nuclear forces were unified as a single force. After cosmic inflation, the unified force separated, giving rise to the laws of nature we know today.</i>
TIME	A hundred-billionth of a yoctosecond (10^{-35} K seconds)	A hundred-millionth of a yoctosecond (10^{-32} K seconds)	1 yoctosecond (10^{-24} K seconds)

all time, space, and energy begins as a point of unimaginable density

cosmic inflation—a dramatic expansion of the Universe

early Universe contains a soup of fundamental particles, such as quarks, gluons, and force-carrying bosons

protons and neutrons—compound particles made of quarks held together by gluon fields—formed nuclei of the lightest elements